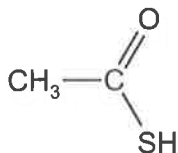
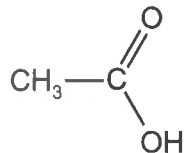


- 3 Thioacetic acid and ethanoic acid both act as weak acids in solution.



thioacetic acid
 $\text{p}K_{\text{a}} = 3.33$



ethanoic acid
 $\text{p}K_{\text{a}} = 4.76$

Which statements are correct?

- 1 H^+ is more easily removed from ethanoic acid because the O–H bond is more polar than the S–H bond.
- 2 The K_{a} value is larger for ethanoic acid because the O–H bond is stronger than the S–H bond.
- 3 In two solutions of the same concentration, thioacetic acid will have a lower pH because the S–H bond is weaker than the O–H bond.

A 1, 2 and 3 **B** 1 and 2 only **C** 2 and 3 only **D** 3 only